AFRICAN LEADERSHIP UNIVERSITY

School of Software Engineering

Research Proposal

Smart Green Hub : A full stack Progressive Web Application for predicting, redirecting and planning surplus plant stock in Kigali nurseries

BSc. in Software Engineering

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List of Acronyms and Abbreviations

ALU African Leadership University

API Application Programming Interface

CSRF Cross Site Request Forgery

DSR Design Science Research

ERD Entity Relationship Diagram

FK Foreign Key

HTTPS Hypertext Transfer Protocol Secure

LCP Largest Contentful Paint

PK Primary Key

PWA Progressive Web Application

REST Representational State Transfer

RFA Rwanda Forestry Authority

RWF Rwandan Franc

SMTP Simple Mail Transfer Protocol

TAM Technology Acceptance Model

UI/UX User Interface and User Experience

UQ Unique Constraint

CHAPTER ONE INTRODUCTION

1.1 Background to the Study

Urban greening has become part of Rwanda's national agenda. The City of Kigali's "My Tree" campaign aims to plant three million trees over five years in homes, roadsides, gardens and recreational zones, and it points residents to nurseries where seedlings are available at about RWF 500 each (allAfrica, 2024a). The Rwanda National Forestry Policy includes an action to develop and implement an urban forest management plan (Ministry of Lands and Forestry, 2018). The need is visible on the ground: Seburanga et al. (2014) found tree cover between 10% and 35% across Kigali neighbourhoods, with plant communities largely dominated by alien species, and with domestic gardens among the most extensive green spaces. Flower and tree nurseries supply the plants for all of these spaces.

The supply side is also changing. The Kigali Modern Tree Nursery in Busanza, Kicukiro, was launched with a stated capacity of more than one million seedlings a year, including ornamental plants, and with greenhouses and automated irrigation (Talibu, 2026). Smaller nurseries operate alongside it, for example in the Kabeza-Kanombe wetland (Kicukiro) and at Nyabugogo (Nyarugenge). These small operators grow flowers and trees on limited land, keep their records by hand, and decide what to plant largely from experience. When a batch is planted in greater quantity than customers buy, the surplus does not disappear. It keeps growing, passes its commercial prime and continues to occupy beds that could hold the next planting cycle.

I visited nurseries at Kabeza and held an initial discussion with their operators. The operators explained that plants which are not sold in time keep growing and stay on their beds, which leaves less land for the next planting cycle. They also said that they keep no structured record of stock, planting dates, sales and expenses, so they cannot easily tell which plants will mature unsold or whether the business is making a profit, and that they have few ways to be found by customers beyond people passing by or word of mouth. Operators also described converting unsold, overgrown plants into charcoal as not available to them. This is consistent with the tighter tree-harvesting rules in Rwanda's 2024 forest law, which requires a permit for harvesting trees and prohibits the cutting, transport, trade or use of poles (immature trees under 20 cm in diameter) without a permit issued for special reasons (allAfrica, 2024b). Whether and how these provisions apply to nursery stock will be confirmed with the Rwanda Forestry Authority (RFA) before the pilot. These discussions

provide an initial requirements basis; the planned needs assessment will establish how often unsold stock occurs and how it is handled now.

Online marketplaces can help a nursery advertise, but advertising does not address the root of the problem, which is that surplus is noticed too late and has nowhere to go. This project therefore does not build an online shop. It proposes Smart Green Hub, a full-stack Progressive Web Application (PWA) that acts as a decision-support and surplus-prevention system for nurseries. It combines batch inventory with sales and expense records, rule-based early warnings about batches likely to become unsold, a surplus board through which schools, sector offices, churches, hotels and community greening groups can request mature plants, and a record of the greening impact. A PWA can be installed from the browser, although installation and offline operation must be deliberately implemented (Mozilla, n.d.). All services will be free to nurseries and institutions.

1.2 Statement of the Problem

First, nurseries have no early warning of surplus. Every batch has a commercial prime window, after which plants become overgrown and harder to sell. Without batch records and sales history in one place, operators discover the problem only when the plants have already outgrown their prime and occupied land for months. Evidence from a related field suggests that awareness is the first barrier: in interviews with 42 foodservice providers in Barcelona, Derqui and Filimonau (2024) identified low awareness of how much waste is generated, limited technological expertise and limited resources as the main reasons businesses do not adopt surplus-reduction apps.

Second, there is no channel for plants that cannot be sold in time. Charcoal is not an outlet that nursery operators report as available, and an overgrown plant cannot return to the seedling stage. Meanwhile, the Kigali tree-planting campaign targets homes, roadsides, gardens and recreational zones (allAfrica, 2024a), and institutions such as schools and community groups need plants for greening. Supply and need exist side by side but are not connected. The circular economy literature describes this kind of connection as keeping resources in use by closing loops rather than discarding them (Geissdoerfer et al., 2017), yet no mechanism exists in this setting for matching a nursery's surplus to a local greening need.

Third, nursery records are fragmented. Stock, planting dates, sales and expenses are kept on paper or in memory, so operators cannot easily see which plants sell, which accumulate or whether the business makes a profit. Existing farm software addresses part of

this: AGRIVI, for example, offers production planning, farm records, input inventory and cost analysis, but its public description is directed at enterprise farms, cooperatives and agri-food companies and does not disclose pricing (AGRIVI, n.d.). Surplus-redistribution apps such as Too Good To Go connect businesses with consumers for surplus food (Too Good To Go, n.d.), but they are built for food, not for living plants with growth stages.

The problem to be addressed is therefore the absence of an affordable, offline-capable tool that predicts which nursery batches will become surplus, redirects unavoidable surplus to institutions that need plants, and uses the nursery's own records to plan future planting. This work introduces such a solution and will evaluate, in a small pilot, whether its functions improve the timeliness of surplus decisions, reduce recording effort and produce completed plant hand-overs.

1.3 Aim and Objectives of the Study

The aim is to design, develop and evaluate a free-to-use PWA that helps flower and tree nurseries in Kigali predict surplus stock, redirect it to institutions that need plants and plan future planting, within a Kigali-based pilot.

1.3.1 Specific Objectives

1. By 16 October 2026, review relevant research and software systems and collect requirements from at least eight nursery operators in Kigali (including Kabeza-Kanombe and Nyabugogo), four institutional plant users and two experts or officials, producing a prioritised requirements list and a baseline description of how nurseries currently keep records and handle unsold stock.
2. By 20 November 2026, implement and test a full-stack PWA covering nursery verification, batch inventory, sales and expense records, rule-based surplus-risk alerts, planting recommendations, a surplus board with requests and fulfilment, impact records and offline synchronisation, with all critical acceptance scenarios passing before the pilot.
3. Between 10 and 17November 2026, pilot the system with a target of five nurseries and three receiving institutions, with the researcher acting as administrator; measure alert accuracy against recorded outcomes, recording time, completed surplus hand-overs and usability, and report the results by 30 November 2026.

1.4 Research Questions

1. What record-keeping, surplus-handling and plant-sourcing difficulties do nursery operators and institutional plant users in Kigali experience in the current process?
2. How can a full-stack PWA implement rule-based surplus-risk alerts, surplus redirection and offline-capable record keeping with clear role boundaries and traceable decisions?
3. What do alert accuracy, recording time, completed surplus hand-overs and user feedback show during the pilot, and how useful and easy to use do participants find the system?

1.5 Project Scope

The research will focus on small and medium flower and tree nurseries in Kigali, including the Kabeza-Kanombe wetland area in Kicukiro, Nyabugogo in Nyarugenge and other parts of the city, together with institutions in Kigali that use plants for greening. The pilot target is five nurseries and three receiving institutions, rather than a country-wide rollout. The needs-assessment and pilot samples may overlap. Recruitment targets are planning assumptions; actual participation and any shortfall will be reported.

The system will support three authenticated roles: nursery operator, receiving institution and administrator. The researcher will operate the administrator workflow during the pilot. Public visitors will see only a directory of verified nurseries and aggregate impact figures. Registration and all core features will be free for every role within this capstone. The system will not collect payment information, charge fees, process transactions or integrate a payment gateway. When a nursery offers surplus plants free or at a reduced price, the price is shown on the listing and any payment is agreed and settled directly between the parties.

Smart Green Hub is not an online shop. It has no shopping cart, checkout, retail product search, payment processing or delivery tracking. The nursery directory only shows the name, location and contact details of verified nurseries so that institutions and visitors can find them. Hand-over of plants is arranged between the nursery and the institution outside the system; the system records the request, the decisions and the confirmed hand-over.

The technical scope includes a React and Tailwind CSS frontend built as a PWA with a service worker and an IndexedDB offline store, a Node.js/Express API, Aiven for MySQL, private storage for verification documents and batch photos, email and web-push notifications, and a Leaflet map based on OpenStreetMap. Surplus prediction is rule-based: a

scheduled job projects each batch's unsold quantity from its stock, its prime-window end date and the recent sales rate of the species, and assigns a Low, Medium or High risk level. The operator always confirms the action to take, and an administrator verifies nurseries and institutions. Machine-learning models, photo-based growth detection, SMS or USSD access, mobile-money payments, delivery logistics, native mobile applications and any guarantee that surplus will be taken up are excluded. Internet access is required for registration, verification, surplus requests and decisions; recording batches, sales and expenses is designed to work offline and synchronise later. The interface will be available in English and Kinyarwanda.

1.6 Significance and Justification

For nursery operators, the platform would provide an organised record of batches, sales and expenses, an early signal about batches that are likely to remain unsold, and a route to place mature plants before they occupy land that is needed for the next cycle. The finance records give operators a direct view of income, costs and profit, which matters because economic motives were the main driver of surplus-app adoption in the study by Derqui and Filimonau (2024).

For schools, sector offices, churches, hotels and community groups, the surplus board would give a single place to find plants nearby, free or at a reduced price, for greening activities that support the Kigali tree-planting campaign (allAfrica, 2024a). For the City of Kigali and other greening partners, the impact records would show how many plants were redirected, to which districts, and how much nursery land was cleared for new planting. The capstone would also provide evidence on whether rule-based early warning and surplus redirection are useful enough to justify continued operation, and on whether a PWA with offline recording is workable for small nurseries.

1.7 Research Budget

Table 1 presents a planning ceiling in Rwandan francs. Existing equipment, campus connectivity and open-source software will be used where available. Hosting and email amounts are contingency allowances rather than supplier quotations; paid services will be used only if existing resources cannot support the pilot. Participants will not fund these costs.

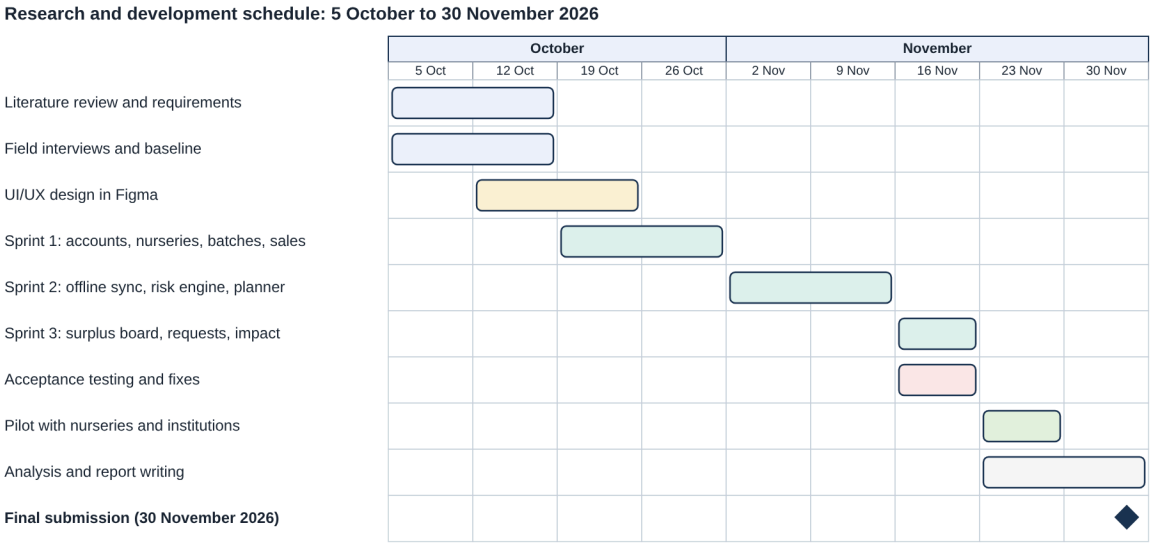
Table 1 Estimated research budget

Item	Basis	RWF
Internet and mobile data	15,000 × 2 months	30,000
Participant travel and local logistics	Field visits to nurseries across Kigali, needs assessment and pilot	40,000
Hosting, storage and email reserve	Only if existing resources are insufficient	40,000
Printing and materials	Consent forms and test materials when needed	5,000
Contingency	10% of RWF 115,000	11,500
Total planning ceiling		**126,500**

1.8 Research Timeline

Figure 1 covers 5 October to 30 November 2026, the final submission date for the capstone. Literature review and requirements collection take place in the first two weeks and are followed by interface design, which overlaps the start of the first implementation sprint. Three implementation sprints run from 19 October to 20 November, with acceptance testing during the last sprint. A one-week pilot follows, and analysis and final reporting are completed by 30 November 2026. Because the schedule is short, the scope is limited to the functions in Section 1.5 and the sample sizes are small.

Figure 1 Research and development timeline



The schedule runs for nine weeks. Shaded bars show planned activity.

Source: *https://drive.google.com/file/d/1VryuvXMhe_-eitVwme6MOQSqnPTZBMm6/view?usp=sharing*

Nursery batch. A group of plants of one species planted on the same date and managed together, with a quantity, a stage and a prime-window end date.

Maturity stage. The growth stage of a batch: seedling, growing, ready or overgrown. A batch that is no longer in use is marked cleared.

Prime window. The period during which a ready batch is at its best commercial condition. It ends on the prime-window end date, after which the batch is treated as overgrown.

Surplus risk. A Low, Medium or High level calculated from the projected unsold share of a batch at the end of its prime window.

Surplus listing. An offer by a nursery of a stated quantity from a batch, free or at a reduced price, published on the surplus board.

Surplus board. The list and map of active surplus listings that verified receiving institutions can browse.

Receiving institution. A verified school, sector office, church, hotel or community group that requests surplus plants for greening.

Impact record. A record created when both parties confirm a hand-over, showing the quantity delivered, the planting site and the land cleared.

Rule-based prediction. A calculation from fixed, documented rules and configurable thresholds, not a trained machine-learning model.

Offline synchronisation. The process by which records created on a device without a connection are stored locally and sent to the server when a connection returns, without duplication.

CHAPTER TWO LITERATURE REVIEW

2.1 Introduction

This targeted review examines urban greening and nursery supply in Kigali, surplus-redistribution and farm-management software, and the technical basis for offline-capable web applications. Sources were identified through title and keyword web searches and checked against publisher or DOI records, official product pages and technical documentation. Search terms included ‘Kigali urban forestry’, ‘Rwanda forest law’, ‘food waste reduction app’, ‘farm management software’, ‘circular economy’ and ‘progressive web app offline’. The review draws on 14 sources: five scholarly works and nine policy, news, product or technical sources. Selection prioritised relevance to the proposed workflow and explicit limitations. It is a focused proposal review, not an exhaustive systematic review or a claim to have searched every indexed database.

2.2 Conceptual Review

Surplus in a nursery is a timing problem. A plant batch has a growing period and a prime window, and unsold stock does not wait; it changes stage while it occupies land. The central concepts of this project follow from that. Prediction means estimating, from stock, remaining prime-window days and recent sales, how much of a batch is likely to remain unsold. Redirection means placing that surplus with users who need it. Planning means using the nursery’s own sales history to decide how much to plant next.

A decision-support system in this sense does not replace the operator’s judgement. The platform calculates a risk level and suggests an action, and the operator confirms or ignores it. This matters because the inputs are simple records entered by the operator, and an alert is only as good as those records. The prediction rules are therefore deliberately transparent, so that an operator or a supervisor can see why a batch was flagged.

Redirection draws on the idea of the circular economy, which Geissdoerfer et al. (2017) describe as a regenerative system that minimises resource input and waste by slowing, closing and narrowing resource loops. For a nursery, the loop that matters is the one that moves mature plants to a planting site instead of leaving them to occupy a bed. The intended outcomes are fewer unsold batches, faster clearing of land for new cycles and a recorded contribution to greening. Adoption, record quality and institutional demand will influence

those outcomes, so the evaluation will measure workflow performance and user experience rather than attributing environmental benefit directly to the software.

2.3 Historical Background of the Research Topic

2.3.1 Origins and Early Context

Green space in Kigali has grown mainly through private gardens, cultivated forests and urban woodlots, and its composition reflects the city's rapid urbanisation and differences between neighbourhoods (Seburanga et al., 2014). Plant supply followed the same pattern: many small producers sold plants to local buyers, with little shared information about what was in stock or what was in demand. National policy recognised weaknesses in planting material, noting poor-quality tree seed that was not well adapted to agro-ecological zones and proposing to strengthen the national tree seed centre (Ministry of Lands and Forestry, 2018). Policy attention therefore focused on seed quality and forest cover, while the commercial and record-keeping side of small nurseries received much less attention.

2.3.2 Key Milestones

The 2018 forestry policy set out actions that included an urban forest management plan (Ministry of Lands and Forestry, 2018). In 2024, a new forest law replaced the 2013 law and extended the permit requirement to all tree harvesting regardless of area (allAfrica, 2024b), and the City of Kigali announced its three-million-tree "My Tree" campaign (allAfrica, 2024a). In mid-2026, the Kigali Modern Tree Nursery was launched with a capacity of more than one million seedlings a year (Talibu, 2026). In parallel, digital surplus platforms matured elsewhere: Too Good To Go, founded in Copenhagen in 2016, reports operating in 21 countries with 180,000 business partners (Too Good To Go, n.d.), and researchers have begun to study why businesses do or do not adopt such apps (Derqui & Filimonau, 2024).

2.3.3 Historical Gaps and Transition to the Current Study

The sources show strong policy and demand for urban greening, expanding plant supply and mature surplus-redistribution platforms for food. They do not show a tool that helps a small nursery foresee unsold stock, nor a mechanism that routes that stock to local greening users. The proposed contribution is the design and evaluation of that combination

for Kigali nurseries. It does not depend on claiming that inventory software, farm records or surplus-redistribution apps are new inventions.

2.4 Theoretical Review

2.4.1 Foundational Theories and Framework

Three frameworks guide the project. First, the circular economy (Geissdoerfer et al., 2017) frames surplus plants as a resource to be redirected rather than waste. It is applied cautiously: the platform records plants moved to planting sites but does not measure lifecycle environmental benefit.

Second, Design Science Research (DSR) provides the research framework. An information-system artefact is designed to address a practical problem and is evaluated for utility (Hevner et al., 2004). The project will move from stakeholder requirements to design, implementation, demonstration and pilot evaluation. The proposed relationship is that earlier and clearer information about stock and sales supports better surplus decisions, and that a simple offline-capable tool is more likely to be used.

Third, the Technology Acceptance Model (TAM) proposes that perceived usefulness and perceived ease of use influence whether people accept an information technology (Davis, 1989). The pilot questionnaire will therefore ask nursery operators and institutional users to rate usefulness and ease of use on a five-point scale. The model does not predict long-term adoption, and a small pilot cannot establish it.

2.5 Overview of Existing Systems

AGRIVI and Too Good To Go are selected because both are operational platforms whose official descriptions cover part of the proposed workflow: farm records and inventory on one side, surplus redistribution on the other. Their descriptions provide evidence of advertised functionality; they do not independently prove effectiveness. Table 2 separates documented capabilities from the Kigali-nursery requirements that remain to be evaluated.

2.5.1 Strengths and Weaknesses of Existing Systems

Table 2 Comparison of relevant software platforms

System	Documented strengths	Fit limitation for this study
AGRIVI	Crop and production planning, farm and field records, input inventory, cost analysis by field, crop and hectare, and mobile and offline data collection (AGRIVI, n.d.).	Aimed at enterprise farms, cooperatives and agri-food companies; pricing is not disclosed. The reviewed page does not describe nursery batch maturity windows, surplus prediction or redirecting stock to local institutions.
Too Good To Go	Businesses list surplus and customers reserve and collect it through the app; reports 120 million registered users and more than 500 million meals saved (Too Good To Go, n.d.).	Built for surplus food. It does not model living plants with growth stages or forecast stock, and its reported markets are in Europe, North America and Asia Pacific.

The comparison identifies a contextual fit gap, not proof that these systems lack configurable tools. A large platform may offer broader features, while a purpose-built tool may suit small Kigali nurseries better through simple records, Kinyarwanda, offline use and local matching. The capstone must therefore test whether those local benefits justify the effort of building and introducing a new tool.

2.6 Review of Related Works

Derqui and Filimonau (2024) interviewed 42 foodservice providers in Barcelona about Too Good To Go. Economic motives were the prime driver of adoption and environmental motives were secondary; low awareness of waste generation, lack of technological expertise and limited resources were the main demotives, and the authors recommend simplified technology. This supports including finance records that show the economic benefit, alerts that raise awareness of waste early, and a simple interface. The setting, hospitality food businesses in Spain, differs from Kigali nurseries, so the findings cannot indicate the size of any effect here.

Seburanga et al. (2014) combined interviews, field surveys and remote sensing to describe Kigali's tree cover and green-space composition, finding that tree cover ranged from 10% to 35% and that alien species dominated plant communities. This establishes demand and variety in the city's planting, but the study does not examine nursery operations or surplus.

Geissdoerfer et al. (2017) review the circular economy as a sustainability paradigm. It motivates the redirection function, but it is a conceptual review and does not provide a method for matching nursery surplus to institutional need.

The technical sources show that the intended offline design is feasible. Mozilla's guide describes service workers that intercept requests and serve cached responses, background synchronisation that runs when connectivity returns, and push notifications (Mozilla, n.d.), and Workbox provides service-worker libraries that simplify caching and offline behaviour (Google, n.d.). Browser support for these features varies, so the design also synchronises when the app detects that connectivity has returned, and the behaviour will be tested on the phones used in the pilot.

2.6.1 Summary of Reviewed Literature

Table 3 Literature synthesis and design implications

Source	Evidence and implication	Limit relevant to this study
Derqui & Filimonau (2024)	Interviews with 42 providers; economic motives and awareness drive adoption of surplus apps. Motivates finance records, early alerts and a simple interface.	Food businesses in Spain; no estimate of effect for Kigali nurseries.
Seburanga et al. (2014)	Survey of Kigali tree cover and green space. Motivates demand for plants across neighbourhoods and institutions.	Does not study nurseries or surplus.
Geissdoerfer et al. (2017)	Circular economy review. Motivates redirecting surplus as a resource.	Conceptual; gives no matching method.

Source	Evidence and implication	Limit relevant to this study
Hevner et al. (2004)	Design Science Research; guides artefact creation and evaluation.	Provides a method, not evidence that this artefact will succeed.
Davis (1989)	Perceived usefulness and ease of use influence acceptance. Guides the pilot questionnaire.	Does not predict long-term adoption.
Mozilla (n.d.); Google (n.d.)	Service workers, caching, background sync and push support offline-first design.	Support varies by browser and must be tested on pilot devices.

2.7 General Comments

The review supports three design decisions: keep records simple and tied to a visible economic benefit, make surplus visible early, and provide a route to place it. It leaves untested whether a rule-based early warning and a surplus board work for small Kigali nurseries. Within the sources reviewed, no system combines nursery batch maturity tracking, surplus prediction and matching of surplus plants to local institutions, and this is the gap the project addresses. The project evaluates whether the combined process reduces recording effort, flags unsold batches in time and produces completed hand-overs, with free participation for nurseries and institutions.

CHAPTER THREE SYSTEM ANALYSIS AND DESIGN

3.1 Introduction

The project will use Design Science Research with mixed-methods evidence and iterative full-stack development. Quantitative measures will capture alert accuracy, recording time and hand-over outcomes; interviews and observation will explain usability difficulties. The design uses a modular web application and a relational data model. The prediction is rule-based, so model training datasets and machine-learning baseline sections are not applicable.

3.2 Research Design and Development Method

3.2.1 Participants and Sampling Strategy

Purposive sampling will recruit nursery operators who grow flowers or trees for sale in Kigali, institutions that use plants for greening, and experts or officials familiar with urban greening. Operators already visited at Kabeza will be invited first, and they will be asked to introduce other operators (snowball referrals) so that Kabeza-Kanombe, Nyabugogo and at least one other part of Kigali are covered. Institutions will be recruited through schools, sector offices, churches, hotels and community groups near the participating nurseries. Experts or officials will be approached at the City of Kigali and the RFA. The needs assessment targets eight nursery operators, four institutions and two experts or officials. The pilot targets five nurseries and three institutions, with the researcher as administrator.

Nursery operators must run a nursery in Kigali, be willing to share stock and sales information, use or be able to use a smartphone, and give informed consent. Recruitment will seek variation in nursery size, plant type and location rather than a statistically representative sample. Participants may join both phases. Non-participation will have no consequence, and actual sample sizes and any missing responses will be reported.

3.2.2 Data Collection and Analysis

The needs assessment will use a short questionnaire on current record keeping, unsold stock, ways of handling it and sourcing of plants, followed by semi-structured interviews with approximately four operators, two institutions and the two experts or officials. Interview prompts will ask participants to describe a recent batch that was not sold in time and how the

decision was made. An agreed checklist will turn those findings into acceptance scenarios, and they will also be used to confirm the default prime-window lengths and thresholds.

Three evaluation activities follow. First, alert accuracy will be tested by back-testing: for historical batches recorded by participating nurseries (target at least 10 where records exist), the rules will be run as at a date several weeks before the prime window ended, and the result will be compared with what actually happened to the batch. Second, recording effort will be compared by asking each pilot operator to record the same ten representative stock and sales entries in their usual method and in the app, and comparing median times. Third, during the pilot, timestamped records will capture alerts, operator actions, listings, requests, decisions and confirmed hand-overs, and a five-point questionnaire based on perceived usefulness and ease of use (Davis, 1989) and short exit interviews will collect user feedback.

Analysis will use counts, proportions, medians and recurring interview themes. Back-test and pilot figures will be reported separately, with denominators and missing data stated. A small purposive sample and the absence of a control group limit causal and population-wide conclusions.

3.2.3 Agile Development and Research Ethics

Requirements will be collected from 5 to 16 October 2026 and the interface designed in Figma from 12 to 23 October. Three implementation sprints will follow: 19 to 30 October for accounts, nurseries, batches, sales and expenses; 2 to 13 November for offline synchronisation, the risk engine and the planting planner; and 16 to 20 November for the surplus board, requests, impact records and deployment. Each sprint will end with a demonstration and backlog review, and acceptance testing runs alongside the last sprint. The one-week pilot (23 to 27 November) and reporting follow development rather than being treated as additional coding sprints.

Participation will be voluntary, with separate consent for research participation, for publishing the nursery in the public directory, and for sharing contact details with an institution. Only necessary data will be collected, documents and photos will be private, access will follow role and purpose, and de-identified data will be used in reporting. Photographs will not include people’s faces. Rwanda’s Data Protection and Privacy Office administers registration for data controllers and processors and requires authorisation for storing personal data on servers outside Rwanda (Data Protection and Privacy Office, n.d.); the institution and supervisor will confirm responsibilities and the hosting location before real

personal records are used. The proposed retention period for pilot data is 90 days after the final report unless a documented requirement or a participant request requires a different lawful period. Withdrawal and correction requests will be handled through the administrator. During development, synthetic data will be used. Before the pilot, the researcher will confirm with the RFA how forest-law provisions apply to nursery stock and to any suggested use of overgrown plants.

3.2.4 Metrics for Evaluation

Table 4 distinguishes intended success criteria from results that will be observed. Thresholds are proposed acceptance targets, not completed findings. Environmental benefit and income change are outside the short pilot's primary outcomes.

Table 4 Evaluation measures and proposed targets

Measure	Calculation or method	Proposed criterion
Alert precision	Batches flagged Medium or High that ended with unsold stock ÷ all batches flagged, in the back-test.	At least 70%; report numerator and denominator.
Alert recall	Batches that ended with unsold stock and were flagged at least 14 days before the prime window ended ÷ all batches with unsold stock.	At least 70%; report missed batches.
Recording efficiency	Median time to record ten stock and sales entries in the app versus the usual method.	At least 25% less time; report both medians.
Offline sync correctness	Offline records synchronised without loss or duplication ÷ all offline records in test scenarios.	100% on agreed scenarios before pilot.
Decision traceability	Surplus decisions with actor, timestamp and reason ÷ all decisions.	100% in the evaluated cases.

Measure	Calculation or method	Proposed criterion
Completed hand-overs	Requests confirmed as handed over by both nursery and institution.	Observe; seek at least three.
Plants redirected	Total quantity and number of planting sites recorded in impact records.	Report only; no target.
Task completion	Operators completing registration and adding a batch without researcher help ÷ those attempting.	At least 80% during observed sessions.
Usefulness and ease of use	Median rating on a five-point scale for perceived usefulness and ease of use.	At least 4/5; report response count and comments.

3.3 System Analysis

3.3.1 Functional Requirements of the System

FR01 Free platform access. The system shall provide registration, verification, batch management, surplus listings and requests without charging any role, collecting payment information or processing transactions. It shall not provide a shopping cart, checkout, retail product search or delivery tracking.

FR02 Registration and verification. The system shall allow a nursery operator or a receiving institution to register with an email address and confirm it through a single-use link. A nursery or institution shall not publish or request anything until an administrator approves it after reviewing the submitted details. The administrator role shall be assigned through controlled provisioning, and users shall not be able to grant themselves any role.

FR03 Nursery profile and directory. A verified nursery shall maintain a profile with its name, district, sector, address description, map location and contact details. The public directory and map shall show only approved nurseries whose operators have consented to publication.

FR04 Batch management. A nursery operator shall create, update and close plant batches with species, quantity, planting date and unit price. The system shall compute the expected ready date and the prime-window end date from species defaults that the operator

may adjust, shall move a batch through the stages seedling, growing, ready, overgrown and cleared, and shall keep a record of every stock adjustment with its reason.

FR05 Sales and expenses. A nursery operator shall record sales against a batch and record expenses by category. The system shall reduce batch stock on each sale, reject a sale larger than the available stock, and calculate income, expenses and profit or loss by period, species and batch.

FR06 Rule-based surplus-risk evaluation. The system shall evaluate every active batch each night and on demand. For each batch it shall calculate the sales rate over the lookback period, the days left until the prime-window end, the projected unsold quantity and the risk ratio, and shall assign a Low, Medium or High level using the thresholds in Table 5. A batch that is past its prime window with stock remaining shall be High.

FR07 Alerts and suggested actions. For a Medium or High batch the system shall create one open alert, notify the operator and suggest an action based on the days left. The operator shall acknowledge, resolve or dismiss each alert, and every change shall be recorded. The system shall never change a price, a listing or a batch automatically.

FR08 Planting recommendations. The system shall suggest a quantity for the next planting of a species from its recent sales, its growing cycle and the stock already in the pipeline, as defined in Section 3.3.3, and shall show the basis of each suggestion. When sales history is insufficient, it shall say so instead of suggesting a quantity.

FR09 Surplus listings. A nursery operator shall create a surplus listing from a batch with a quantity, an offer type (free or discounted), a price where discounted, a pickup note and an expiry date. The server shall reject a listing larger than the available stock and shall publish only listings of verified nurseries.

FR10 Surplus board and requests. A verified receiving institution shall browse active listings by list and map, filter by plant type and district, and request a quantity with a stated planting purpose. A duplicate active request from the same institution for the same listing shall return the existing request. A valid request shall reserve the requested quantity.

FR11 Decision and fulfilment. The nursery operator shall approve or decline each request with an optional reason. A declined or expired request shall release the reserved quantity. After the plants are handed over, both the nursery and the institution shall confirm the hand-over; only then shall the system reduce the batch stock permanently.

FR12 Impact records. When both parties confirm a hand-over the system shall create an impact record with the quantity delivered, the planting site, the district and the land cleared. Nursery, institution and administrator dashboards shall summarise plants redirected, hand-overs completed and land cleared, and the public page shall show aggregate figures only.

FR13 Offline recording and synchronisation. The app shall allow an operator to record batches, stock adjustments, sales and expenses without a connection, store them locally with a device-generated unique identifier, and send them when a connection returns. The server shall ignore a change whose identifier it has already applied, shall flag a conflicting change for the operator, and shall return server-side changes to the device.

FR14 Notifications. The system shall notify users of new requests, decisions, alerts and verification outcomes by in-app message, email and web push where the user allows it. Notifications shall use a persistent outbox with one record per event and recipient, a retry status, and visibility of failures for administrative follow-up.

FR15 Administrator dashboard and rules. The administrator shall review and approve or reject nurseries and institutions, moderate listings, maintain plant species and default prime windows, and edit the risk thresholds, lookback period and planting safety margin. The dashboard shall show registered and verified nurseries and institutions, active batches, open alerts by level, active listings, requests, completed hand-overs and total plants redirected.

FR16 Monitoring and audit. The platform shall preserve timestamped account, verification, batch, stock, alert, listing, request, hand-over and rule-change actions with the actor and reason. Records shall be archived rather than permanently deleted.

FR17 Password recovery. Every account shall have a “Forgot password?” link. After a user submits an email address, the system shall return a neutral response and send a single-use reset link that expires after ten minutes. The user shall set a new password, receive a confirmation and have existing sessions invalidated. Administrators shall not view or assign user passwords.

FR18 Language. The interface shall be available in English and Kinyarwanda, and each user shall be able to choose and change the language. Each user shall see dates, quantities and currency in the Rwandan format.

Table 5 sets out the default rules used by FR06, FR07 and FR08. The thresholds are configuration data that the administrator can change, and they will be checked against the needs-assessment findings and the back-test.

Table 5 Surplus-risk rules and suggested actions

Rule	Definition	Default setting
Sales rate r	Units of the species sold during the lookback period ÷ number of lookback days.	Lookback L = 28 days
Days left D	Prime-window end date minus today; 0 if the date has passed.	Species default, editable
Projected unsold U	max(0, S − r × D), where S is the current stock of the batch.	Calculated
Risk ratio R	U ÷ S.	Calculated
Risk level	High if D = 0 and S > 0, or R ≥ 0.50; Medium if 0.20 ≤ R < 0.50; otherwise Low.	0.20 and 0.50
Suggested action	D > 28: reduce next planting or discount. D from 8 to 28: discount or transplant. D ≤ 7 or overgrown: surplus board or donate.	Day limits 28 and 7
Planting recommendation	Average monthly units sold over the last three months × growing-cycle months × (1 + safety margin) − units already in the pipeline; never below zero.	Safety margin 0.10

3.3.2 Nonfunctional Requirements of the System

NFR01 Performance. During a documented 50-user concurrent test, the warmed API shall achieve a 95th-percentile response time below one second for ordinary reads, excluding photo uploads and the nightly job. Key public pages shall target a Largest Contentful Paint (LCP) below three seconds under a recorded mobile test profile. Cold-start behaviour shall be reported separately.

NFR02 Security. All connections shall use HTTPS. Passwords shall be hashed with a maintained password-hashing library. Authentication shall use secure, HttpOnly cookies, server-side role checks, request validation and Cross Site Request Forgery (CSRF) protection where applicable. Verification and reset links shall be hashed, single-use and time-limited. Login, reset and upload routes shall be rate limited, reset responses shall be neutral, and successful password changes shall invalidate existing sessions.

NFR03 Data and file protection. Uploads shall allow PDF, PNG and JPEG only, with a 10 MB limit, and file signatures and sizes shall be checked. Verification documents shall be stored outside public access and retrieved only through authorised, short-lived access. Public responses shall exclude phone numbers, documents and review notes unless the owner has consented.

NFR04 Usability and access. Layouts shall be mobile first and also support desktop screens, with large touch targets, labelled controls with icons, keyboard navigation and clear error messages in English and Kinyarwanda. The app shell and public pages may be cached, but authenticated API responses and private documents shall not be cached by the service worker.

NFR05 Reliability and maintainability. The pilot shall target 99% measured availability, with outages reported rather than hidden. Daily database backups and a restore exercise shall be completed before pilot launch. Versioned migrations, automated critical-path tests, error logging and environment-based secrets shall support maintenance.

NFR06 Offline resilience. Locally stored records shall survive closing the browser and restarting the device. When connectivity returns and the app is open, synchronisation shall start automatically and shall report how many changes were applied, skipped or in conflict.

3.3.3 Proposed System Model

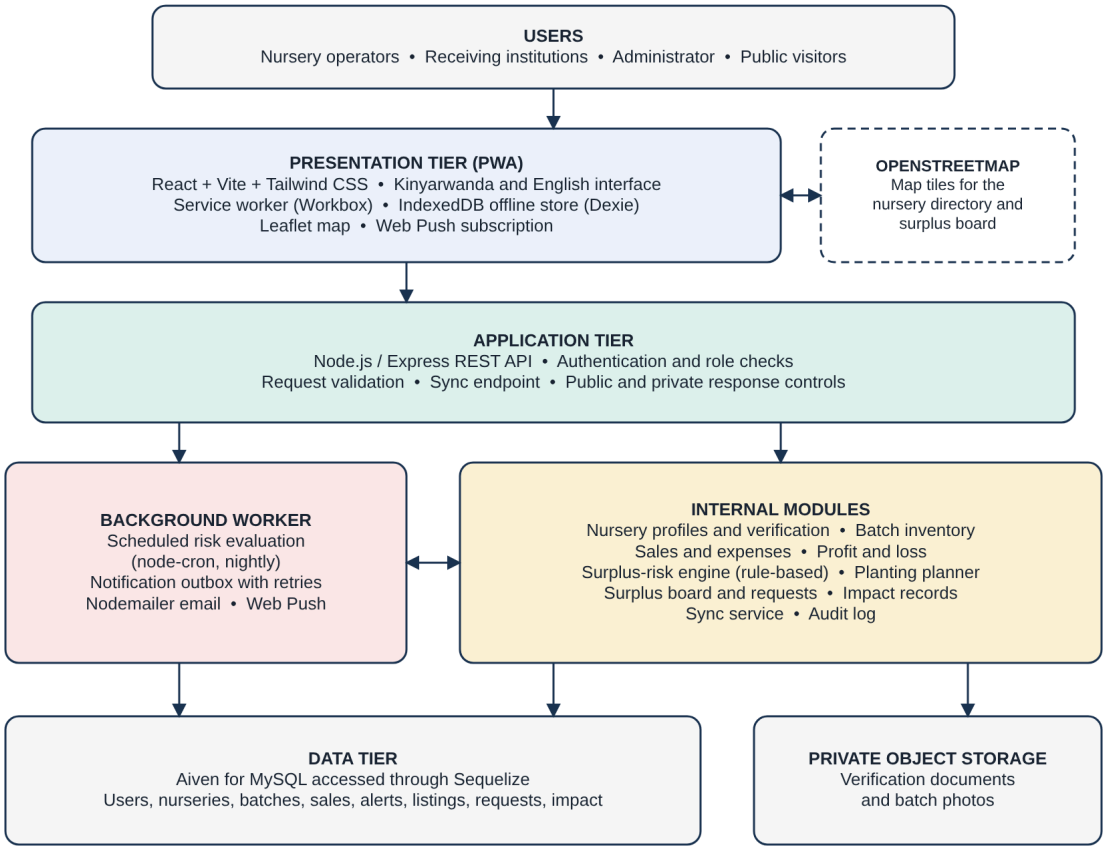
The model separates presentation, application and data responsibilities. React renders the public and role-specific interfaces, including the nursery dashboard, the surplus board and the map. A service worker caches the app shell and public pages, and an IndexedDB store holds records created offline. Express enforces access and workflow rules and exposes the sync endpoint. Aiven for MySQL stores relational records. A risk and planning module applies the rules in Table 5; a surplus module manages listings, requests and hand-over confirmations; an impact module records outcomes; and a sync module applies offline changes idempotently. A background worker runs the nightly risk evaluation and sends queued notifications with retries. These are modules of one deployable backend, not separately operated microservices, and none changes a price, listing or stock without an operator's confirmation. Figure 2 shows the full-stack arrangement.

3.4 System Design

Figures 2 to 12 are original designs for the proposed system. The architecture, three database views, use cases, class structure, workflows, risk logic and sequence diagrams share the same actors and terminology. The accompanying Smart_Green_Hub_Diagrams.drawio file contains an editable page for each figure from Figure 1 to Figure 12.

3.4.1 Full Stack Architecture

Figure 2 Full stack system architecture



All predictions are rule-based decision support. Operators confirm every action and administrators verify accounts.

The application tier controls data and file access. A background worker runs the nightly risk evaluation and sends notifications.

Source: https://drive.google.com/file/d/1VryuvXMhe_-eitVwme6MOQSqnPTZBMm6/view?usp=sharing

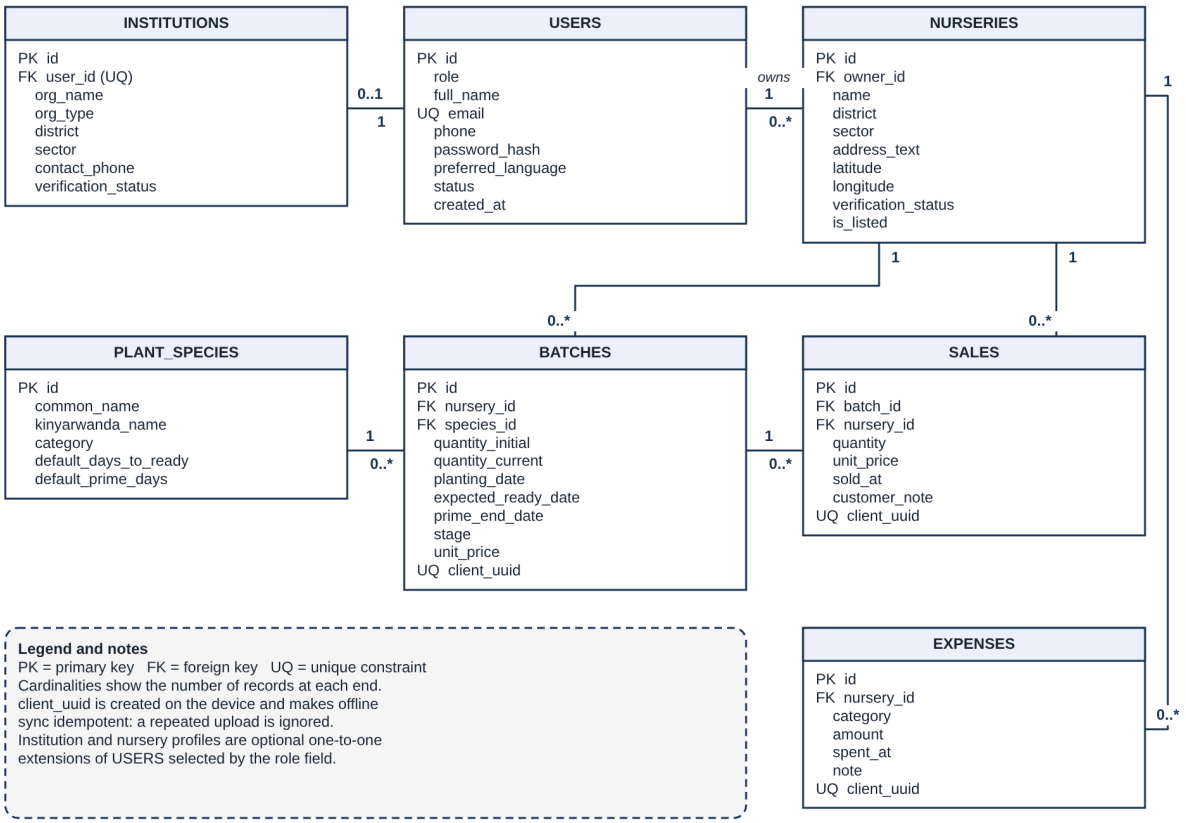
3.4.2 Database Design

Figures 3 to 5 are entity relationship diagrams (ERDs): three views of one logical schema with 15 entities. Dashed boxes are entities defined in another view. An institution profile is an optional one-to-one extension of USERS, and one user account may own several nurseries; the role field controls which is valid. NURSERIES.owner_id identifies the responsible operator account. BATCHES stores the stage and the prime-window end date,

and each of BATCHES, SALES and EXPENSES carries a device-generated client_uuid with a uniqueness constraint so that a repeated offline upload is ignored.

A surplus listing belongs to one batch and one nursery. A request belongs to one listing and one institution, and the pair (listing, institution) is unique. Reserving, releasing and permanently reducing stock happen in database transactions so that a batch can never be offered or handed over in a larger quantity than it holds. An impact record references exactly one request. Risk thresholds and lookback periods are stored as configuration data in RULE_SETTINGS, where a row with no species is the default rule. Notifications use a unique event key per event and recipient. Supporting authentication-token tables and standard created and updated timestamps are omitted from the diagrams for readability.

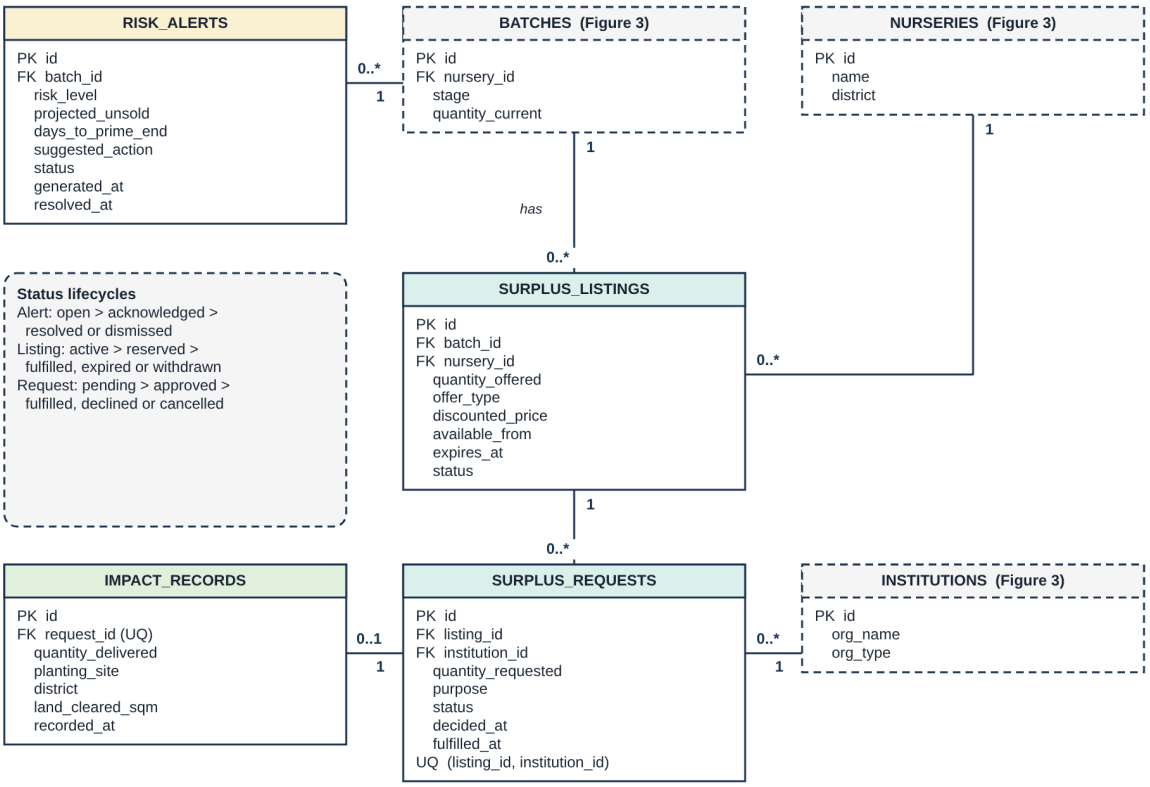
Figure 3 Entity relationships for identity, inventory and finance



Cardinalities describe the number of records at each end. This view covers accounts, nurseries, batches, sales and expenses.

Source: https://drive.google.com/file/d/1VryuvXMhe_-eitVwme6MOQSqnPTZBMm6/view?usp=sharing

Figure 4 Entity relationships for risk alerts, surplus and impact

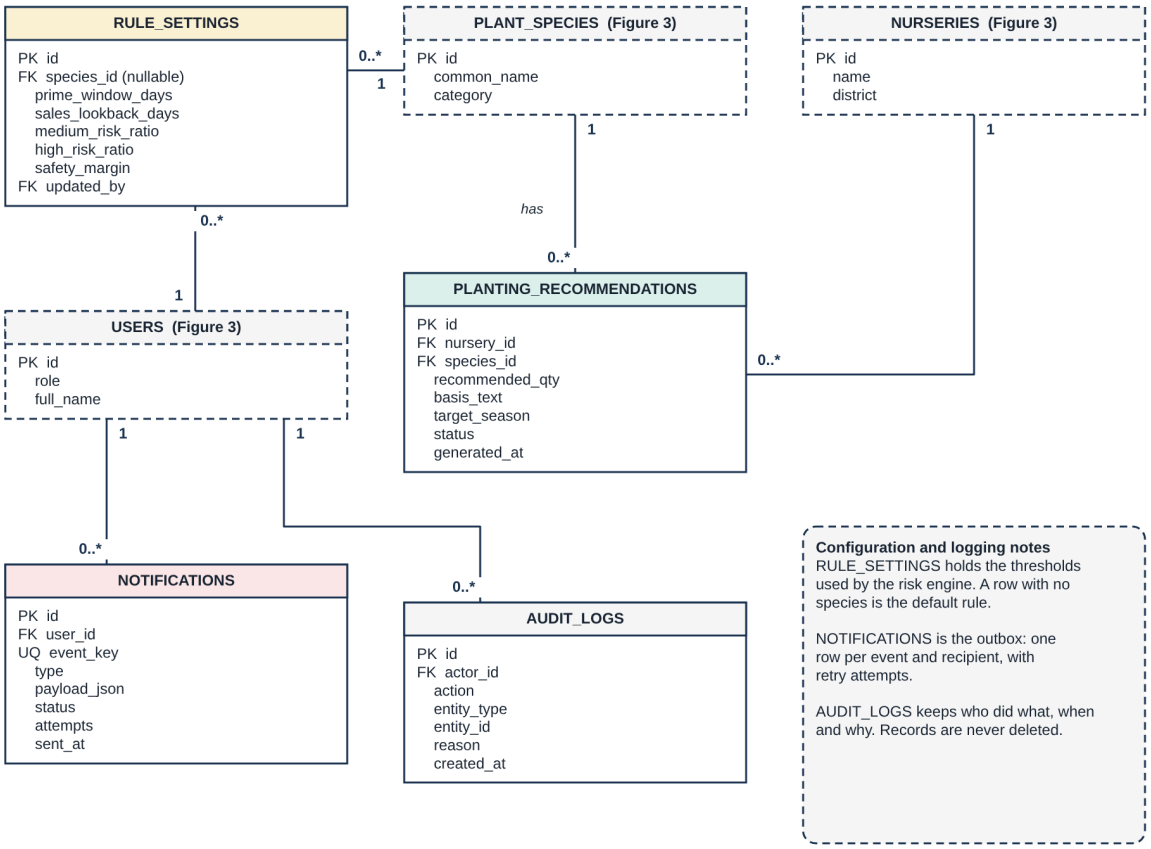


This view shares BATCHES, NURSERIES and INSTITUTIONS with Figure 3 and adds alerts, listings, requests and impact records.

Source: https://drive.google.com/file/d/1VryuvXMhe_-eitVwme6MOQSqnPTZBMm6/view?usp=sharing

03_ERD_risk_surplus_impact.

Figure 5 Entity relationships for rules, planning, notifications and audit



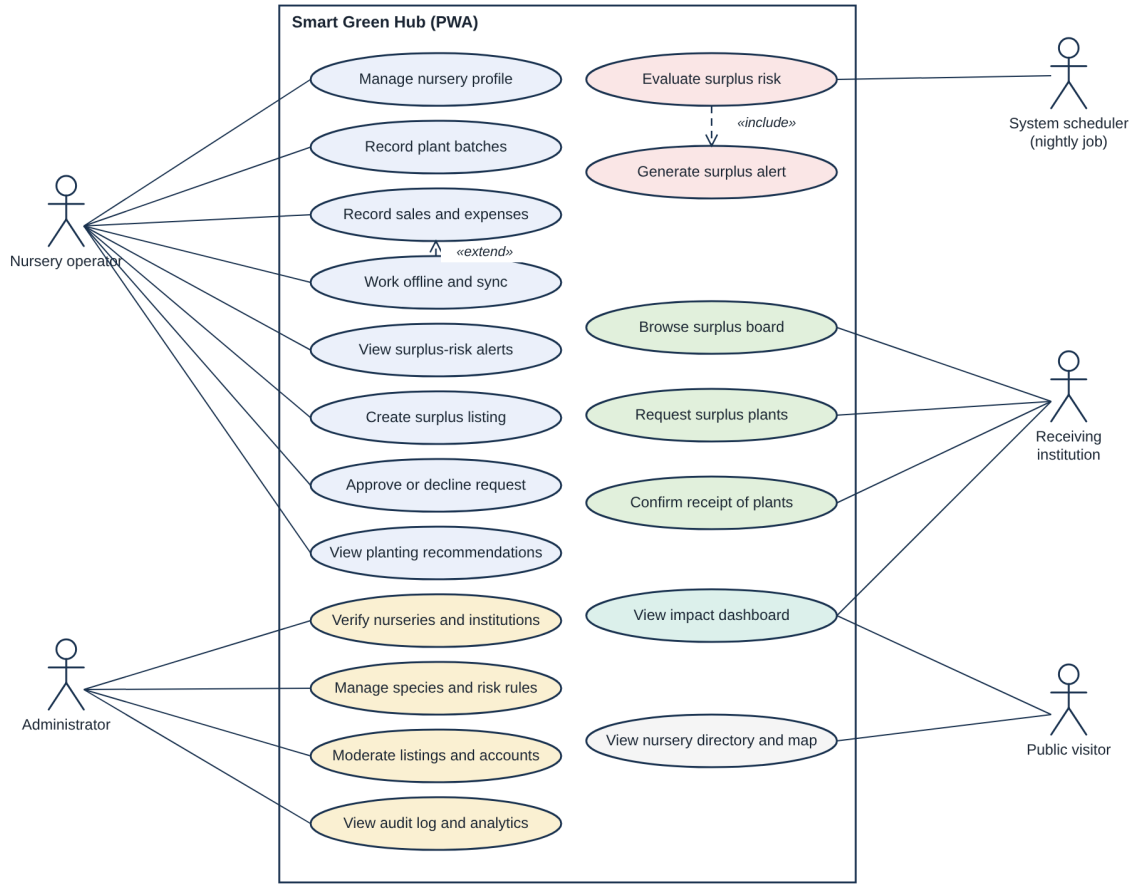
This view covers configuration, planting recommendations, the notification outbox and the audit log.

Source: https://drive.google.com/file/d/1VryuvXMhe_-eitVwme6MOQSqnPTZBMm6/view?usp=sharing

04_ERD_planning_rules_notifications.

3.4.3 Use Case Diagram

Figure 6 Actor responsibilities and system use cases



Alerts shown to operators are produced by the scheduled job; operators confirm every action. Work offline and sync extends the recording use cases.

The nursery operator manages stock and surplus decisions. The administrator verifies accounts and maintains rules. The scheduled job produces alerts.

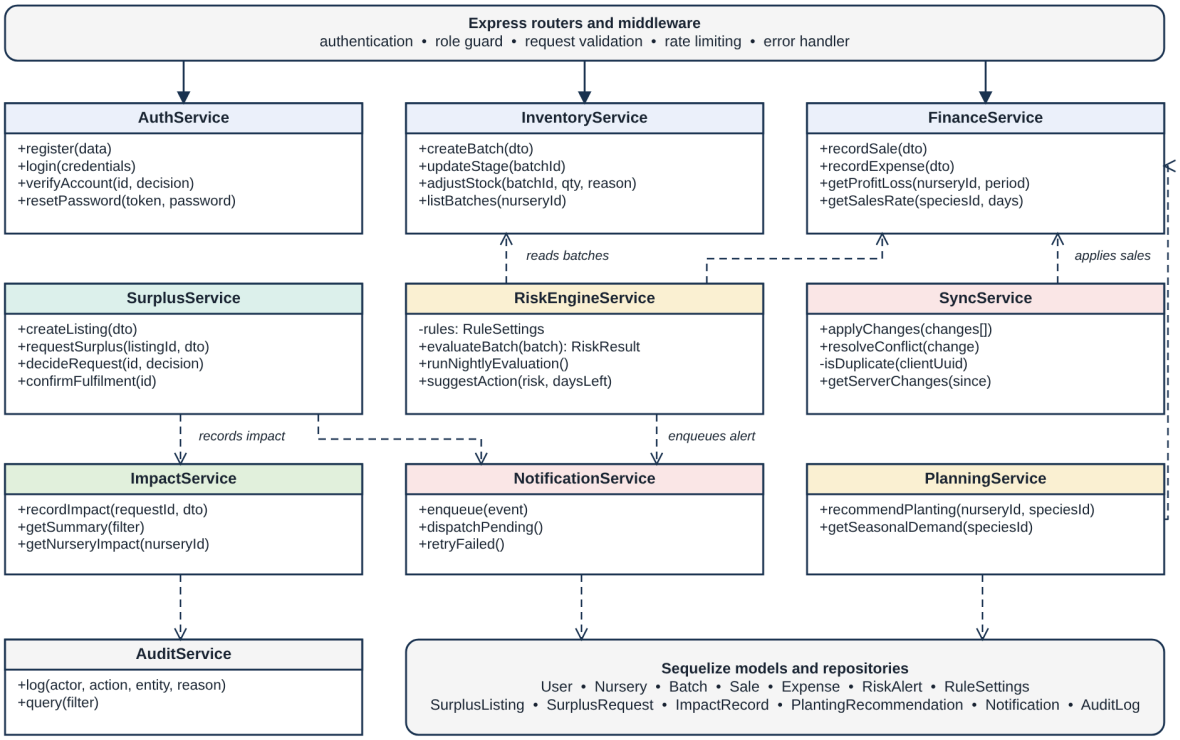
Source:

https://drive.google.com/file/d/1VryuvXMhe_-eitVwme6MOQSqnPTZBMm6/view?usp=sharing

05_Use_cases.

3.4.4 Class Diagram

Figure 7 Backend service classes and dependencies



Dashed arrows show 'uses' dependencies between the main services. Every service reads and writes through the Sequelize models, and every state-changing service writes to AuditService.

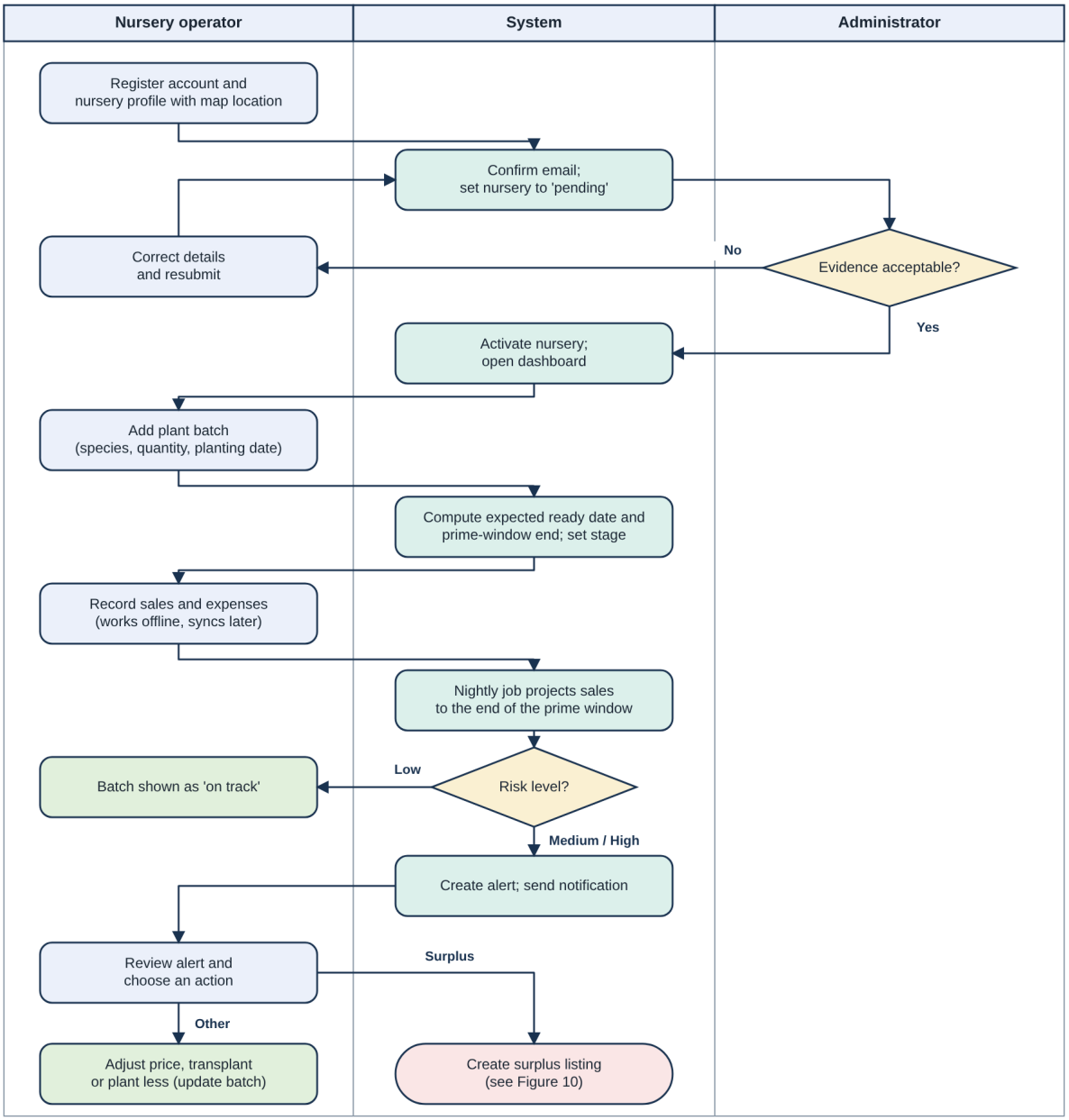
The services encapsulate business operations and access the relational entities through Sequelize models and repositories.

Source:https://drive.google.com/file/d/1VryuvXMhe_-eitVwme6MOQSqnPTZBMm6/view?usp=sharing

06_Service_classes.

3.4.5 Nursery Onboarding and Batch Workflow

Figure 8 Nursery registration, batch recording and alert workflow



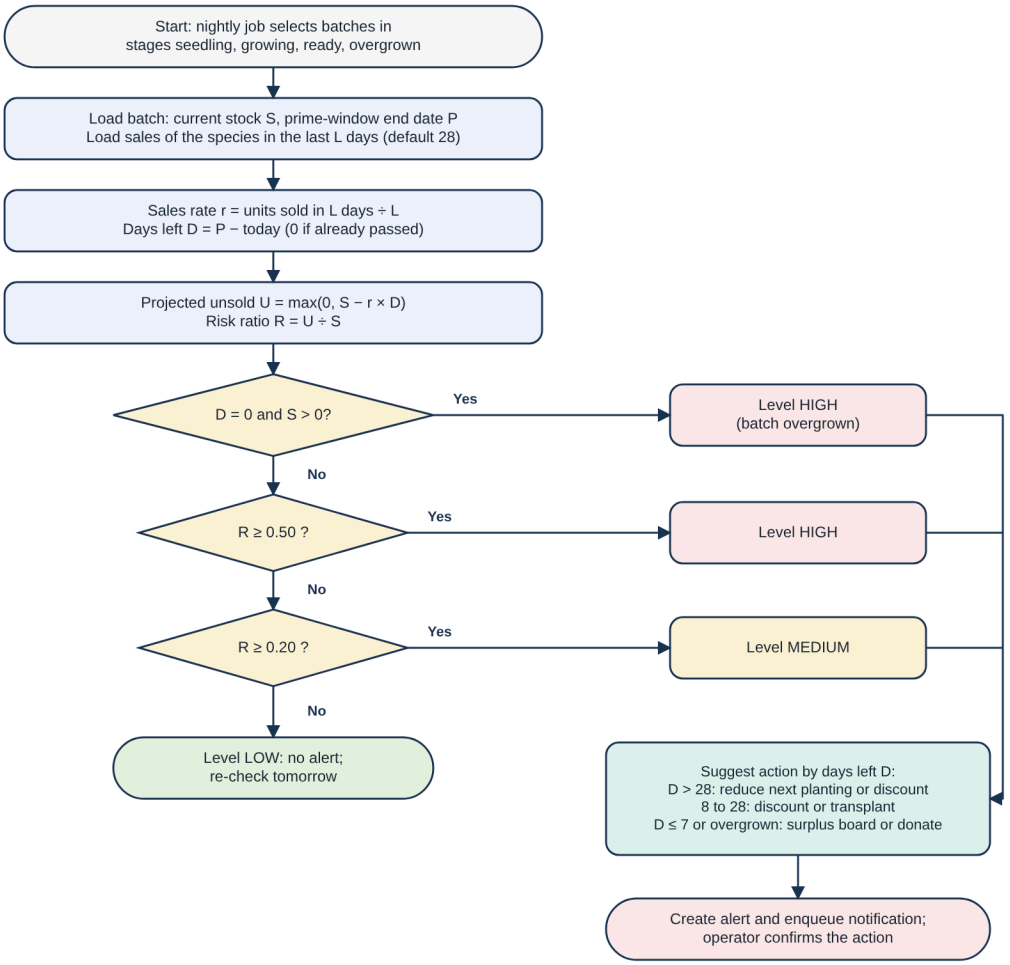
Verification precedes batch recording. The nightly job creates an alert for Medium and High batches, and the operator chooses the action.

Source: https://drive.google.com/file/d/1VryuvXMhe_-eitVwme6MOQSqnPTZBMm6/view?usp=sharing

07_Onboarding_batch_workflow.

3.4.6 Surplus-Risk Evaluation Logic

Figure 9 Rule-based surplus-risk evaluation



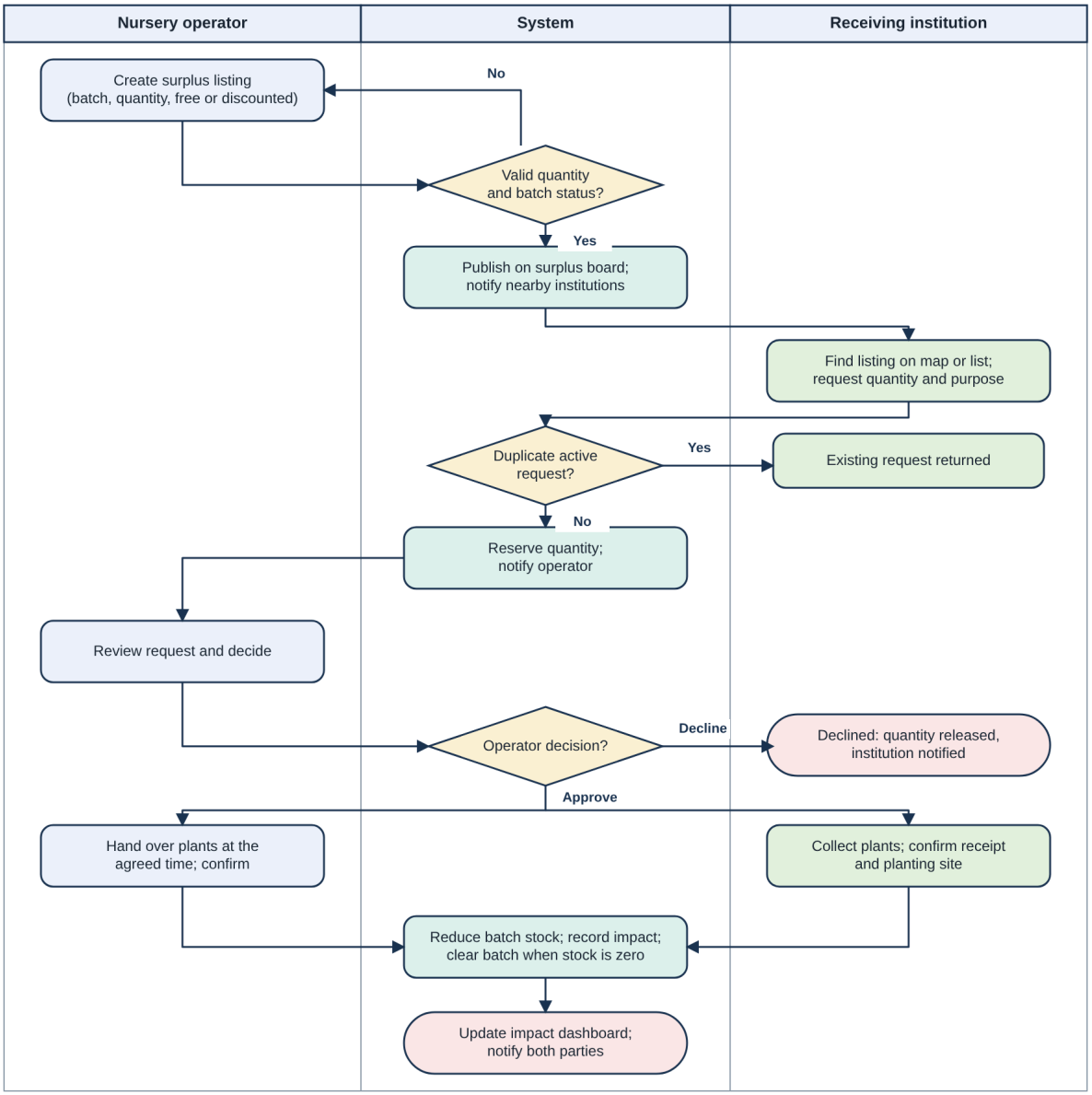
The calculation uses stock, days left and the recent sales rate. It produces a risk level and a suggested action, never an automatic change.

Source: https://drive.google.com/file/d/1VryuvXMhe_-eitVwme6MOQSqnPTZBMm6/view?usp=sharing

08_Risk_evaluation_logic.

3.4.7 Surplus Redirection Workflow

Figure 10 Surplus listing, request, hand-over and impact



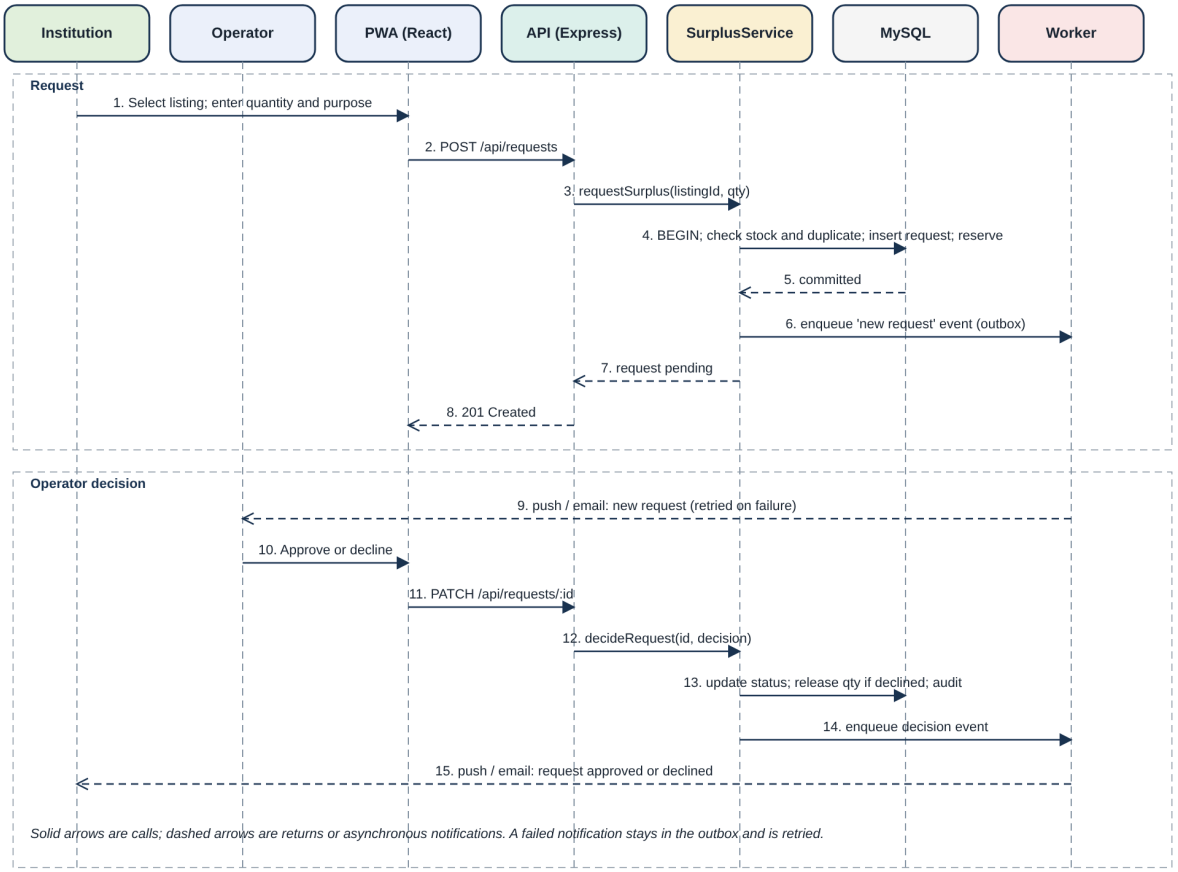
The server validates the listing and prevents duplicate requests. The operator decides, and stock is reduced only after both parties confirm the hand-over.

Source:https://drive.google.com/file/d/1VryuvXMhe_-eitVwme6MOQSqnPTZBMm6/view?usp=sharing

09_Surplus_redirection_workflow.

3.4.8 Surplus Request Sequence Diagram

Figure 11 Surplus request and decision sequence



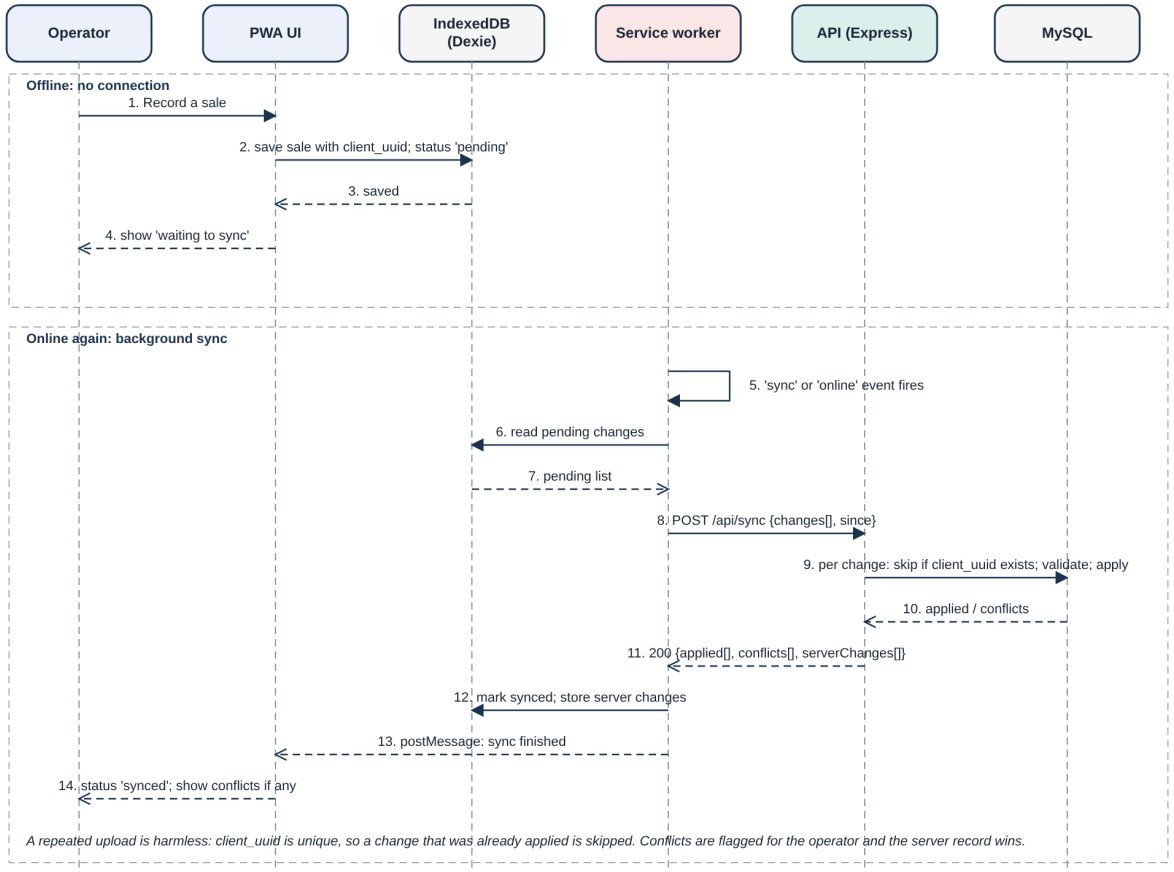
The request path reserves stock in one transaction and queues notifications. Delivery failures stay in the outbox and are retried.

Source: https://drive.google.com/file/d/1VryuvXMhe_-eitVwme6MOQSqnPTZBMm6/view?usp=sharing

10_Surplus_request_sequence.

3.4.9 Offline Synchronisation Sequence Diagram

Figure 12 Offline recording and synchronisation sequence



Changes are saved locally with a device identifier and sent when the connection returns. A repeated upload is skipped.

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11_Offline_sync_sequence.

3.5 Development Tools

Table 6 Development tools and intended use

Area	Tools	Purpose
Frontend	React, Vite, Tailwind CSS, Workbox, Dexie (IndexedDB), Leaflet	Responsive PWA interface, offline store and service worker, nursery directory and surplus map, and role-based dashboards.
Backend	Node.js LTS, Express, node-cron	REST API, access control, risk evaluation, planting recommendations, sync endpoint and workflow services.
Persistence	Aiven for MySQL, Sequelize	Relational records, constraints, transactions and migrations.
Files	Private object storage	Verification documents and batch photos.
Notifications	Nodemailer and SMTP, web push	Email and push notices for alerts, requests and decisions, with retry tracking.
Testing	Jest, Supertest, Playwright, Lighthouse	Business-rule, API and browser tests; offline, performance and accessibility checks.
Design and delivery	Figma, draw.io, Git and GitHub	Interface design, diagrams, version control and deployment workflow.

Dependency versions will be pinned at implementation using maintained compatible releases. The project will use the smallest hosting arrangement that supports the pilot. Aiven for MySQL will provide the managed relational database, while frontend and API hosting will be selected after checking resource limits and data-handling requirements. No service is assumed to provide free capacity or guaranteed uptime. Draw.io will maintain editable technical diagrams.

A complete UI/UX prototype is being designed in Figma for the public landing page and nursery directory, the sign-in and registration flow, the Nursery Dashboard, Batch Inventory and Add Batch form, Sales and Expenses, Surplus Alerts, Planting

Recommendations, the Surplus Board with map and request form, the Receiving Institution dashboard, the Administrator verification and rules screens, the Impact Dashboard and Account Settings. Together, these screens demonstrate the principal user flows for the nursery operator, receiving institution, administrator and public visitor roles. The prototype uses a consistent mobile-first dashboard system, large touch targets, icons that accompany labels, and an English and Kinyarwanda language switch.

3.6 Future Work and Recommendations

Once the system has collected enough real batch and sales records, the rule-based prediction can be compared with a machine-learning forecast, provided that the data, governance and evaluation are adequate and the operator keeps the final decision. Photo-based growth-stage detection could reduce manual stage updates. Later phases may add SMS or USSD access for basic phones, mobile-money payments for discounted listings, delivery partners, QR plant passports that show planting dates and care instructions, and links to City of Kigali and RFA planting programmes. The legal and environmental suitability of other uses of overgrown vegetation, such as mulch or compost, should be reviewed with the RFA before any such use is suggested. Machine-learning forecasting, payments, delivery and native applications are not included in the present capstone.

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